



Software Requirements Specification (SRS) & System Design

Comprehensive Architectural & Functional Feature Inventory

Project Title:	RENTZY — Real Estate, Student Housing & Co-Living Platform	Document Version:	2.0 (Complete Edition)
Technology Stack:	React.js (Vite), Tailwind CSS, Spring Boot, Capacitor (PWA/Mobile), Leaflet GIS, WebSocket/STOMP	Date of Release:	July 2026

1. Executive Summary & System Overview

RENTZY is a next-generation, cloud-native real estate rental and shared housing marketplace designed to eliminate traditional rental friction, brokerage fees, and safety risks. Built as a Progressive Web Application (PWA) with native mobile readiness via Capacitor, RENTZY serves four primary user personas: **Tenants/Seekers**, **Property Owners/Landlords**, **Packers & Movers**, and **System Administrators**.

The platform incorporates GIS mapping, algorithmic smart compatibility matching for roommates, peer-to-peer expense splitting, real-time WebSocket communication, and gender-customized UI theming for student accommodations.

2. Functional Requirements by Module

MODULE 1 User Authentication & Identity Management

- **Multi-Channel Authentication:** Supports secure Email/Password onboarding with BCrypt password hashing and **Google OAuth 2.0 Single Sign-On (SSO)** via @react-oauth/google and Capacitor native sign-in.
- **Role-Based Access Control (RBAC):** Dynamic routing and permissions for Tenants, Owners, Packers & Movers, and Admins.
- **Landlord KYC & Fraud Prevention:** Government ID upload workflow (Aadhaar/PAN/Passport) reviewed by admins. Approved landlords receive a prominent  badge on all their listings.
- **Account Recovery:** Secure OTP/Email-based password reset and credential recovery.

MODULE 2 Property Discovery & GIS Search Engine

- **Multi-Category Hubs:** Dedicated portals for *Flats & Apartments* (1RK to 4BHK+), *PGs & Hostels*, *Co-Living Spaces*, *Commercial Offices*, and *Warehouses*.
- **Granular Search Filters:** Real-time filtering by Budget, City/Locality, Configuration, Furnishing Status, Mess/Meals availability, and Tenant Preferences (e.g., *Bachelors Men*, *Bachelors Women*, *Family*).
- **Interactive GIS Leaflet Mapping:** OpenStreetMap/Google tile integration displaying properties as interactive map pins with floating price tags and instant preview popups. Reverse geocoding for precise address rendering.
- **Wishlist & Search Alerts:** One-click bookmarking with instant backend synchronization and automated search alert triggers when new matching inventory is published.

MODULE 3 Student PG & Hostel Portal (Gender-Themed UI)

A specialized student accommodation portal featuring tailored aesthetic UI themes to enhance trust and user experience:

-  **Girls PGs & Hostels (Rose/Pink Theme):** Styled with a soft rose palette (border-pink-200, rose badges) emphasizing women-only safety features: 24/7 CCTV surveillance, biometric campus access, female wardens, and hygienic home-cooked meals.
-  **Boys PGs & Hostels (Sky/Blue Theme):** Styled with a clean sky blue palette (border-blue-200, blue badges) highlighting high-speed WiFi, recreational lounges, laundry services, and zero brokerage.
- **Interactive Portal Switcher:** Dynamic tab navigation allowing students to switch between Girls, Boys, and All PGs, fetching live verified listings directly onto the portal page with URL parameter synchronization.

MODULE 4 Roommate Finder & Smart Compatibility Matching

- **Community Request Boards:** Users post personal roommate requests specifying budget, move-in date, location, and flat sharing preferences. Dedicated tabs for Women-Only and Men-Only roommate communities.
- **Algorithm-Powered Smart Match Engine:** Calculates a weighted percentage compatibility score between users based on 8 key lifestyle criteria:

Compatibility Factor	Options / Scale	Algorithm Weight
Smoking Habits	Non-Smoking vs Smoking Okay	15%
Drinking Habits	Non-Drinking vs Drinking Okay	12%
Dietary Preference	Vegetarian / Vegan vs Non-Vegetarian	10%
Sleep Schedule	Early Bird vs Night Owl	10%
Cleanliness Level	Immaculate vs Moderate / Casual	10%
Pet Comfort	Pets Welcome vs No Pets	8%
Budget & Location Alignment	Target Rent & Locality Overlap	25%
Gender Alignment	Target Gender Preference Match	10%

MODULE 5 Rent & Expense Splitting Suite

- **Shared Household Groups:** Flatmates create digital household groups (e.g., "*Flat 402 Buddies*") and invite members via shareable invite codes or QR links.
- **Itemized Bill Logging:** Record household bills categorized by Rent, Groceries, Electricity, WiFi, Water, Dining, and Maid/Maintenance. Support for equal or custom percentage splits.
- **Automated Debt Simplification:** Real-time ledger displaying net peer-to-peer balances (Green for owed amount, Red for owing amount). One-click "*Settle Up*" transaction logging.

MODULE 6 Real-Time Messaging, Chat & Notifications

- **WebSocket / STOMP Chat Engine:** Asynchronous real-time messaging between tenants, landlords, and matched roommates without page reloading.
- **Contextual Inquiry Routing:** Clicking "Message Owner" on a listing automatically opens a chat session pre-populated with property reference cards and inquiry text.
- **Notification Bell & Alerts:** Live header bell icon with unread badge counters tracking lease updates, visit confirmations, chat messages, and rent payment reminders.

MODULE 7 Landlord Portfolio & Relocation Movers Suite

- **Step-by-Step Listing Creation:** Multi-step form with Cloudinary image upload (auto-compression and WebP delivery) and interactive GPS map pin drop for latitude/longitude.
- **Occupancy Lifecycle Management:** Landlords toggle property status between *ACTIVE* (Vacant/Available) and *RENTED* / *ARCHIVED*.
- **Integrated Packers & Movers Marketplace:** Tenants relocating can discover verified moving companies, filter by local shifting or inter-city transport, and submit inventory size (1RK to 3BHK+) for instant competitive price quotations.

MODULE 8 Governance, Administration & Legal Compliance

- **System Admin Control Console:** User moderation (ban/unban accounts), Landlord KYC document verification queue, and property listing audit/takedown tools.
- **Two-Way Rating & Review System:** Transparent rating mechanism where tenants review landlords/properties and owners review tenants/roommates.
- **Legal & Support Center:** Comprehensive FAQs, IT Act/GDPR compliant Privacy Policy, Terms of Service, Cookie Policy, and floating AI customer support widget.

3. Non-Functional Requirements (NFRs) & System Architecture

NFR Category	Implementation Details & Architecture
Responsive Design & PWA	Built using React.js (Vite), Tailwind CSS, and glassmorphism styling. 100% responsive across mobile phones, tablets, and desktop displays. Progressive Web App capabilities for home-screen installation.
Native Mobile Ready	Capacitor framework integration (@capacitor/core, @capacitor/haptics) enabling tactile haptic feedback and ready for compilation into native Android (.apk/.aab) and iOS apps.
High Performance & CDN	Cloudinary media transformation engine delivering auto-compressed, responsive image formats (f_auto,q_auto:good). Spring Boot backend caching (@Cacheable, @CacheEvict) for sub-100ms API response times.
Security & Compliance	Stateless Spring Security JWT authentication, BCrypt password hashing, CORS origin whitelisting, CSRF/XSS protection, and secure HTTPS/REST endpoints.